Keshav Rajput

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Data engineer / data scientist, new grad. MS Computer Science, NYU Tandon, May 2026. PySpark, SQL, pandas, R.
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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2026
Master of Science (MS) in Computer Science GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi 2020 - 2024
Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics CGPA 8.19/10

Coursework: Big Data, Principles of Database Systems, Information Visualization, Foundations of Data Science, Statistical Inference, Machine Learning

PROJECTS

NYC AutoDDG | Python, PySpark, pandas, PyArrow, Ollama | github.com/happyc0der/nyc-autoddg 2026

- Built a single-pass PySpark profiler (types, statistics, semantic types, data-quality flags) for 100 NYC Open Data datasets: 4.8M rows in 2.3 minutes, down from 12; checked against a pandas oracle with pytest.
- Benchmarked Spark against pandas on an M4 Pro and Databricks serverless over Parquet: crossover at 1.7M rows, near-linear scaling to 8 cores, pandas 14x faster on small files.
- Had a local qwen3:14b model write dataset descriptions from the profiles; 97% of checkable claims matched the data, and search Recall@1 rose from 0.605 to 0.728.

TrafficFlowBench (in progress) | Python, PyTorch | github.com/happyc0der/trafficflowbench 2026

- Scored 0.889 on the Kaggle TrafficFlowBench (IEEE Big Data Cup 2026) public leaderboard against a 0.553 baseline, up from 0.674 on the first submission; final ranking pending.
- Built the four-task pipeline: missing cells from LightGBM residual models plus a PyTorch U-Net, LightGBM time-series queue forecasters trained out of core on 56M rows with causal features, and an exact NNLS origin-destination solver.
- Found a window-selection regularity that lifted queue-onset IoU from 0.57 to 0.93 and shared it on the forum; one script rebuilds the submission bit-for-bit in about two hours.

Kaggle S6E9, EV purchases | Python, LightGBM 2026
github.com/happyc0der/kaggle-s6e9-ev-purchases

- Finished 287th of 3,575 teams (top 9%) in Kaggle Playground S6E9 (ROC AUC): private 0.94545 against a winning 0.94602, out-of-fold 0.94648.
- Engineered features with nested target encodings over the synthetic generator's artifacts on 668,665 rows with 20-fold cross-validation; blended LightGBM, CatBoost and a 15-seed MLP in logit space.
- Judged 192 experiments by paired DeLong hypothesis tests; label resampling showed a perfect model would score 0.9459 +/- 0.0003, so the remaining gap is label noise.

Biohub cell tracking (Finished) | Python, PyTorch, NumPy, SciPy, polars 2026

Builds on Igor Zharov's public notebook (Apache-2.0); the harness, classifiers and experiments are mine.

- Scored 0.952 on the Kaggle Biohub Cell Tracking public leaderboard on free Kaggle GPU, against 0.947 for the public notebook it builds on; finished 591st of 3,947 teams (top 15%).
- Built an offline harness that runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos; it reproduces the notebook's score to within 0.00011.
- Trained a 3D CNN on 134 annotated cell divisions to prune false branches: predicted +0.0047 on a pre-registered holdout, measured +0.005 on the leaderboard twice.

COMPETITIONS

- **Money Talks 2, AI x Fintech Hackathon (Block Convey), Money Operations track**, 2026. Solo entry Ledger, a forecast agent that checks every claim against the CRM; 94% recall and 0 false alarms on 235 deals. Result pending.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Parquet, time-series forecasting, statistics (hypothesis testing, regression), econometrics (fixest), Streamlit, Altair, Matplotlib

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes to hawk-py
- Events head, Finnexia (finance club), IIIT Delhi; organized "The Critical Bid"